

CellSim: A Domain-Driven Software Framework for Stochastic Tumour Evolution Simulation

Introduction and Related Work (Draft)

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Contents

1	Introduction	2
2	Related Work	3
2.1	Mathematical Models in Cancer Biology	3
2.2	Bioinformatics, Software Engineering, and the Reproducibility Chal- lenge	5
2.3	Towards Integrated Frameworks: CellSim as a Proof of Concept . . .	6

1 Introduction

Cancer is not a single disease but a heterogeneous family of conditions sharing a common hallmark: the progressive and irreversible corruption of the cellular mechanisms that govern proliferation, differentiation, and death [12, 1]. Tumour development is inherently a multi-scale, stochastic process that simultaneously spans the molecular level—where individual DNA strand breaks, gene silencing events, and protein–protein interactions occur—the cellular level—where fitness differences between clones are resolved through competition—and the tissue level—where the tumour microenvironment, immune surveillance, and spatial constraints shape the evolutionary landscape [10, 3]. This complexity is not merely descriptive: it has direct clinical consequences. Interventions that succeed *in vitro* routinely fail *in vivo*, and population-level epidemiological trends do not reliably predict individual-patient outcomes, precisely because the stochastic nature of early oncogenesis is systematically underestimated [1].

In response to this challenge, the past six decades have witnessed an accelerating effort to bring mathematical and computational tools to bear on cancer biology [9, 1]. The aspiration is compelling: if cancer follows deterministic or stochastic laws at the cellular level, then those laws should be amenable to formal analysis, and models derived from them should generate testable, quantitative predictions. The results, however, have been mixed. As Chauviere and colleagues noted in their critical appraisal of the field, “*over-optimistic estimations about [mathematical oncology’s] ability have created unrealistic expectations*” [1]. May articulated the underlying methodological trap with even greater precision: models are frequently constructed with “*an excruciating abundance of detail in some aspects, whilst other important facets of the problem are misty*” [5]. A model that is mathematically sophisticated yet biologically misspecified does not merely fail to predict—it may actively mislead experimental design and clinical decision-making.

The failure modes of incorrect models are not merely theoretical. A deterministic ordinary differential equation (ODE) system may capture average tumour growth dynamics while systematically missing the stochastic rare events—a single cell acquiring a second loss-of-function allele in a tumour suppressor gene, for instance—that are responsible for the transitions between disease phases [2, 6]. Cell–cell communication networks introduce frequency-dependent fitness effects into the competitive landscape of the tissue: the selective advantage of an immunosuppressive phenotype is not constant but depends on the current composition of the tumour microenvironment [10, 8]. These phenomena demand stochastic, agent-level modelling

frameworks rather than mean-field approximations.

Despite the recognised need, the translation from sound mathematical theory to reproducible, extensible, and clinically applicable software remains incomplete. The bioinformatics community has documented this gap extensively: technical debt, lack of software engineering discipline, and absent continuous integration practices compromise the reproducibility of computational findings and prevent their adoption in clinical workflows [4]. The problem is not a shortage of mathematical ideas—the theoretical foundations are deep and well-established—but rather a shortage of software artefacts that implement those ideas with the rigour, transparency, and modularity that translational research demands.

This paper presents CELLSIM, a domain-driven, open-source software framework for stochastic tumour evolution simulation, as a step towards closing this gap. CELLSIM is designed as a proof of concept for the intersection of rigorous mathematical modelling, biologically grounded cellular mechanics, and production-quality software engineering. The remainder of this section motivates the three principal contributions of the framework. Section 2 reviews the relevant literature across the three axes—mathematical models, biological mechanisms, and software engineering practice—and positions CELLSIM within that landscape.

2 Related Work

2.1 Mathematical Models in Cancer Biology

Mathematical modelling in oncology has a history of over sixty years. Bibliometric analysis confirms that the field has grown substantially in both volume and interdisciplinary reach, with strong co-authorship networks spanning mathematics, physics, and clinical oncology [9]. The diversity of modelling approaches reflects the multi-scale nature of the underlying biology: deterministic models based on ordinary or partial differential equations have been used to describe macroscopic tumour growth dynamics and treatment response [1]; agent-based models (ABMs) capture spatial heterogeneity and clonal competition at the cellular level; and stochastic frameworks address the inherently probabilistic nature of molecular-level events.

Among stochastic models, two foundational contributions stand out. Moran’s 1958 model of random processes in genetics [6] established the mathematical framework for gene-frequency dynamics in a finite, fixed-size population. In Moran’s formulation, instead of the simultaneous generational replacement assumed by the Wright–

Fisher model, births and deaths occur as individual, asynchronous events: at each step, one individual is chosen uniformly at random to die and is replaced by the offspring of a randomly sampled survivor. This single modification—substituting individual stochastic replacement for synchronous generation turnover—has profound consequences for evolutionary dynamics: the rate of approach to homozygosity is exactly twice that of the Wright model, because the random choice of the individual to die introduces an additional source of variance [6]. Translated to oncology, the Moran process provides a rigorous framework for modelling clonal competition in homeostatic tissue, where cell number is regulated and each division event is coupled to a death event. The probability that a single mutant cell—say, one carrying a loss-of-function allele in *TP53*—ultimately fixes in the tissue, and the expected time to fixation or loss, are directly computable from Moran’s results.

Gillespie’s Stochastic Simulation Algorithm (SSA) [2] addressed a complementary problem: how to perform exact numerical simulation of coupled stochastic reaction systems in continuous time. Gillespie’s key insight was philosophical as much as technical. As he wrote, “*the stochastic formulation of chemical kinetics has a firmer physical basis than the deterministic formulation*”, because the deterministic rate equation is merely the expected-value projection of the underlying master equation [2]. The SSA generates statistically exact trajectories of the master equation without solving it analytically, making it possible to simulate rare molecular events—the very events responsible for somatic mutation and phenotypic switching—with mathematical exactness. This is precisely the regime where ODE-based models are least reliable: the second loss-of-function mutation in a tumour suppressor gene affecting a small number of cells in a large tissue is a rare event that ODE approximations systematically misrepresent.

When cellular fitness is not constant but depends on the current composition of the population—as occurs in the tumour microenvironment, where immunosuppressive cells alter the selective landscape for neighbouring cells [10]—the appropriate theoretical framework is evolutionary game theory. Nowak and Sigmund demonstrated that evolutionary dynamics can be frequency-dependent in a manner that is analytically tractable using replicator equations and evolutionary stable strategy analysis [8]. The implication for tumour modelling is direct: the competitive advantage of a cell with an immunoevasive phenotype is not fixed, but depends on the proportion of such cells already present in the tissue, a feedback that purely kinetic models cannot capture. Nowak’s comprehensive treatment of evolutionary dynamics [7] further established that fixation probabilities and times in finite populations under selection—central quantities for understanding clonal expansion—are analytically

accessible within these frameworks.

Despite this strong theoretical foundation, Hassan and colleagues [3] identify a persistent gap between mathematical models and clinical applicability: most models are developed and validated on *in silico* or cell-line data without external validation against independent clinical cohorts. The combination of machine learning and mechanistic mathematical modelling holds promise for closing this gap, but the technical prerequisites—software frameworks that are both mathematically rigorous and experimentally interoperable—are still largely absent.

2.2 Bioinformatics, Software Engineering, and the Reproducibility Challenge

A second body of literature relevant to this work concerns not the mathematical content of models but their software realisation. The rapid growth of biological data and the proliferation of computational tools have created a secondary crisis: much of the software produced by researchers is unreproducible, undocumented, and impossible to extend or validate independently [4]. Ma and colleagues [4] provide a systematic review of bioinformatics software development practices, identifying technical debt—the accumulated cost of shortcuts taken during development—as the primary obstacle to reproducibility. Their analysis identifies several critical deficiencies: absence of automated testing, lack of version control discipline, hardcoded parameters that prevent systematic sensitivity analysis, and monolithic architectures that resist extension or combination with other tools.

Veneziano and colleagues [12] document the practical consequences of this gap in the context of breast cancer bioinformatics: despite the availability of rich transcriptomic, genomic, and proteomic data, the field lacks software tools capable of integrating multi-omic signals into coherent mechanistic narratives. The gap is not conceptual but infrastructural.

Thomas and Barry [11] argued over two decades ago for a domain-oriented approach to software development in which software abstractions are organised around the conceptual vocabulary of the domain—in this case, cell biology and mathematical genetics—rather than around computational convenience. Their model-driven, domain-oriented programming (DOP) framework anticipates what is now called Domain-Driven Design (DDD) [11]: the principle that the structure of the software should reflect the structure of the domain it models, making the codebase intelligible to biologists, mathematicians, and software engineers alike. This ap-

proach is particularly relevant for scientific computing, where the primary consumer of the software is often a domain expert who needs to reason about model behaviour without reverse-engineering implementation details.

The intersection of the two deficits—missing mathematical rigour in implementations, and missing software quality in mathematical tools—defines the space that the present work aims to address.

2.3 Towards Integrated Frameworks: CellSim as a Proof of Concept

The preceding review identifies a three-way gap at the intersection of cancer biology, stochastic mathematical modelling, and software engineering. Mathematical oncology has produced powerful theoretical tools—the Moran process, the Gillespie SSA, evolutionary game theory—but these tools remain largely disconnected from software artefacts that implement them in a reproducible, extensible, and biologically grounded way. Pugh and colleagues [9] confirm that interdisciplinarity is the defining characteristic of successful mathematical oncology research, yet the software infrastructure required to support multidisciplinary collaboration is systematically underdeveloped.

CELLSIM is designed as a proof of concept for an integrated approach to this problem. It is an open-source, C++20 cellular evolution simulator that adopts a hexagonal (ports-and-adapters) architecture [11], organising all domain logic—cell lifecycle, genetic state, signal integration—in a pure domain layer that depends on no external framework and is independently testable.

The biological model at the core of CELLSIM is grounded in established cancer genetics. Each `AgenticCell` carries structured representations of tumour suppressor genes (*TP53*, *BRCA1*) with biologically constrained state transitions: allelic states progress strictly from wild-type heterozygosity (+/-) to homozygous loss (-/-), consistent with the two-hit model of tumour suppressor inactivation [12]. Two continuous damage accumulators— D_1 for DNA damage and D_2 for immunosuppressive signalling—mediate the transition between six biologically defined cell life stages (BASELINE $\rightarrow$ UNSTABLE $\rightarrow$ UNPROTECTED $\rightarrow$ PRIMER $\rightarrow$ TUMORAL). This staging mirrors the multi-step model of carcinogenesis documented in the literature [1, 12].

The cell lifecycle is implemented as a strictly ordered sequence of biological decisions: intrinsic apoptosis mediated by *TP53* is evaluated before extrinsic immune-mediated apoptosis, which precedes neoplastic development and cytoplasmic remodelling lead-

ing to division. This ordering is not arbitrary but reflects the known hierarchy of cell fate decisions in oncogenesis [10].

From an engineering perspective, CELLSIM implements continuous integration with a dual-compiler matrix (GCC and Clang) and a comprehensive unit-test suite. All architectural decisions are documented as Architecture Decision Records (ADRs), providing a transparent audit trail that is essential for scientific reproducibility and for onboarding contributors from different disciplinary backgrounds.

Critically, the architecture is designed from the outset for extension to the mathematically rigorous stochastic frameworks reviewed in Section 2.1. The Moran process extension will model tumour initiation in homeostatic tissue: when a cell carrying a *TP53* loss-of-function mutation divides and displaces a wild-type neighbour, that is precisely a Moran replacement event, and the probability of clonal fixation is directly given by Moran’s formula [6]. The Gillespie SSA extension will simulate rare somatic mutation events—including the stochastic acquisition of the second *BRCA1* hit—with mathematical exactness, without the approximation errors that ODE-based models incur in the low-copy-number regime [2]. Tau-leaping [2] will provide a computationally tractable approximation for the high-proliferation, large-cell-number regime associated with the Big Bang dynamics of aggressive tumour growth.

Together, these components instantiate the three-way integration that the preceding literature review identifies as necessary but currently absent: a biologically grounded domain model, implemented with production-quality software engineering, that can accommodate the full range of established stochastic mathematical frameworks for tumour evolution.

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